

CHAN YI HENG

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EDUCATION

National University of Singapore

Master of Science in Statistics

Singapore, SG

Aug 2026 – Expected 2027

Singapore University of Social Sciences

Bachelor of Science (Honours) in Mathematics

Singapore, SG

Jan 2022 – Dec 2025

Singapore Polytechnic

Diploma in Banking & Finance

Singapore, SG

Apr 2016 – Mar 2019

EXPERIENCE

Teaching Assistant

Singapore University of Social Sciences

Jan 2026 – Present

Singapore, SG

- Facilitated technical tutorials in Statistical Analysis covering probability theory, asymptotic inference, and classical linear regression.
- Formulated structural review frameworks to ground undergraduate cohorts in deductive mathematical reasoning and problem-solving.
- Guided student execution of formal model diagnostics, emphasizing verification of Gauss-Markov assumptions and residual structures.

Research Assistant

Singapore University of Social Sciences, University Research Office

Oct 2025 – Present

Singapore, SG

- Formulated statistical inference frameworks for experimental psychometric data to quantify treatment effects on latent creativity constructs.
- Developing stochastic models for financial time-series data to capture non-stationarity, temporal dependencies, and volatility dynamics.
- Applying survival analysis and classification techniques to clinical datasets to isolate features driving patient non-attendance.
- Engineered reproducible data pipelines for high-dimensional institutional datasets to ensure mathematical integrity for academic review.

Research Intern

A*STAR, Advanced Remanufacturing and Technology Centre

Nov 2024 – Aug 2025

Singapore, SG

- Architected deep learning models for multi-horizon time-series forecasting under non-stationary demand conditions.
- Implemented a SHAP-based Explainable AI (XAI) framework to quantify localized feature attributions and map non-linear interactions.
- Evaluated contemporary deep learning baselines, optimizing hyperparameter spaces to enhance empirical robustness and performance bounds.

Finance Operations Analyst

Indeed

Sep 2021 – Dec 2021

Singapore, SG

- Audited cross-border financial data systems to ensure high-fidelity data migration and compliance with international reporting frameworks.
- Optimized workflows within Salesforce environments by implementing data validation protocols to minimize transactional errors.

Finance Specialist

Singapore Armed Forces

Aug 2019 – Aug 2021

Singapore, SG

- Managed operational budgets and resource allocations under central government regulatory frameworks.
- Executed internal audits and risk assessments, applying structured methodologies to report compliance metrics to leadership.
- Administered large-scale operational budgets during COVID-19 deployments, maintaining accuracy under highly volatile conditions.

Wealth Management Intern

DBS Bank

Sep 2018 – May 2019

Singapore, SG

- Conducted quantitative operational analyses, leveraging process data to isolate system inefficiencies and formulate optimization strategies.
- Enforced compliance protocols through the systematic validation of Accredited Investor documentation and data control frameworks.

RESEARCH

- Predictive Analytics & Financial Time-Series Modeling** | *Dr. Fan Zengyan* Jun 2026 – Present
- Investigating non-stationary asset dynamics and temporal volatility dependencies within high-frequency financial time-series datasets.
 - Deriving stochastic predictive architectures to model underlying market mechanisms, bypassing the limitations of traditional stationary assumptions.
 - Engineering data ingestion pipelines to process and structure multi-asset institutional streaming data from Bloomberg terminals for rigorous downstream quantitative estimation.
- Stratified Non-Attendance Modeling in Clinical Datasets** | *Dr. Fan Zengyan* Jun 2026 – Present
- Establishing a quantitative framework to isolate the demographic and behavioral covariates driving stratified patient non-attendance in localized clinical settings.
 - Formulating survival analysis and categorical data models to evaluate time-to-event dynamics and map high-variance predictors of non-compliance.
 - Deriving statistical diagnostics to adjust for real-world irregularities, applying heteroscedasticity-consistent estimators and non-parametric variance partitioning.
- Causal Machine Learning for Treatment Heterogeneity Modeling** | *Dr. Fan Zengyan* Oct 2025 – Jun 2026
- Investigated conditional treatment heterogeneity across high-dimensional feature spaces to derive welfare-maximizing assignment rules.
 - Architected a tripartite causal machine learning framework, benchmarking Meta-Learners against Causal Forest Double Machine Learning to isolate individualized treatment effect surfaces.
 - Enforced a rigorous identification protocol utilizing cross-validated propensity scores and deterministic boundary trimming to satisfy overlap assumptions and bound regularization bias.
 - Synthesized localized individualized treatment effect vectors into an unsupervised clustering matrix to extract latent responder archetypes, validating results via an Optimal Policy Tree algorithm.
- Multivariate Time Series Forecasting for Equity Price Prediction** | *Dr. Zhou Tianyi* Jan 2025 – Nov 2025
- Evaluated the out-of-sample empirical stability of multi-horizon (1-, 5-, 20-day) return forecasting frameworks across NYSE-listed equities under non-linear asset dynamics.
 - Conducted systematic comparative evaluations of classical linear estimators against sequential deep learning and contemporary self-attention architectures.
 - Isolated economically meaningful technical signals via ensemble-based XGBoost feature selection pipelines, utilizing Recursive Feature Elimination and permutation importance.
 - Validated the predictive superiority of attention mechanisms in capturing long-range dependencies through rigorous walk-forward backtesting regimes.
- Data-Driven Framework for Enhancing Supply Chain Resilience** | *Dr. Liu Ning* Nov 2024 – Aug 2025
- Investigated the theoretical and empirical performance boundaries of deep learning optimization architectures for stochastic inventory control against classical Predict-Then-Optimize paradigms.
 - Synthesized dilated temporal Convolutional Neural Networks and Multi-Quantile Recurrent Neural Networks to model probabilistic, non-stationary demand distributions.
 - Captured long-range temporal dependencies to optimize direct cost-functions in high-variance multi-SKU simulation environments.
 - Demonstrated significant structural cost reductions and enhanced system resilience relative to traditional heuristic benchmarks.

COMPETENCY

Languages & Scripting: Python, R, SQL, LaTeX
Development & Tools: Git, VS Code, Visual Studio, Jupyter
Spoken Languages: English (Fluent), Mandarin (Native)